

Assignment: Classical Cryptography and Steganography

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1. Scenario: Event Communication Security

A student organization plans to share confidential event information with its core members using a simple classical encryption method. During testing, the message "**MEET AT NOON**" is encrypted using a Caesar Cipher with a fixed shift of **4** positions.

a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.

The Caesar Cipher is one of the oldest and simplest encryption techniques. It is a substitution cipher in which each letter of the plaintext is shifted by a fixed number of positions in the alphabet. In this scenario, every letter is shifted **4 positions forward** while spaces remain unchanged.

The encryption process is shown below:

M → Q

E → I

E → I

T → X

A → E

T → X

N → R

O → S

O → S

$N \rightarrow R$

Therefore,

Plaintext: MEET AT NOON

Ciphertext: QIIX EX RSSR

The encrypted message appears as **QIIX EX RSSR**, which hides the original message from anyone who does not know the encryption rule.

b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

An authorized member who knows that the Caesar Cipher uses a shift of **4** can easily recover the original message by reversing the process. Instead of shifting letters forward, each ciphertext letter is shifted **4 positions backward**.

For example:

$Q \rightarrow M$

$I \rightarrow E$

$I \rightarrow E$

$X \rightarrow T$

$E \rightarrow A$

$X \rightarrow T$

$R \rightarrow N$

$S \rightarrow O$

$S \rightarrow O$

$R \rightarrow N$

After applying the reverse shift to every letter, the original message becomes:

MEET AT NOON

This process is known as **decryption**. Since both the sender and receiver know the same shift value, communication can be carried out successfully. However, the Caesar Cipher provides very little security because there are only 25 possible shift values. An attacker can simply try every possible shift using a brute-force attack and easily discover the original message. Therefore, although the Caesar Cipher is useful for learning the basics of cryptography, it is not suitable for protecting sensitive information in modern communication systems.

2. Scenario: Intercepted Suspicious Message

During routine network monitoring, a cybersecurity intern intercepts the coded message:

"GSRH RH Z HVXIVG"

The message appears to have been encrypted using a simple substitution cipher rather than a complex modern encryption algorithm.

a. Deduce the most likely original message.

The intercepted message is most likely encrypted using the **Atbash Cipher**. The Atbash Cipher replaces every letter with its opposite letter in the alphabet.

The substitutions are:

$G \rightarrow T$

$S \rightarrow H$

$R \rightarrow I$

$H \rightarrow S$

$RH \rightarrow IS$

$Z \rightarrow A$

$HVXIVG \rightarrow SECRET$

After decrypting the entire message, the original plaintext is:

THIS IS A SECRET

b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

Several observations suggest that the Atbash Cipher has been used.

First, the word **RH** appears twice in the ciphertext. When decrypted using Atbash, both occurrences become **IS**, which is one of the most common words in English. Repeating words often provide useful clues during cryptanalysis.

Second, the single-letter word **Z** decrypts to **A**, another common English word. Single-letter words in English are generally limited to "A" and "I", making them useful hints for identifying substitution ciphers.

Third, the final word **HVXIVG** decrypts perfectly into **SECRET**, producing a meaningful sentence.

The Atbash Cipher is a monoalphabetic substitution cipher in which every plaintext letter always maps to the same ciphertext letter. Because of this fixed substitution, patterns in the ciphertext are preserved, making the cipher relatively easy to identify and break through pattern analysis and frequency analysis. Although historically important, the Atbash Cipher offers very weak security and is unsuitable for protecting modern digital communications.

3. Scenario: Secure Messaging System

A company uses a keyword-based encryption technique known as the **Vigenère Cipher** to protect internal communication. An employee encrypts the word "HELLO" using the keyword "KEY".

a. Demonstrate how the encryption process transforms the message step-by-step.

The Vigenère Cipher is a polyalphabetic substitution cipher. Instead of using one fixed shift value, it uses multiple shift values determined by a repeating keyword.

Step 1: Write the plaintext.

Plaintext:

HELLO

Step 2: Repeat the keyword until it matches the length of the plaintext.

Keyword:

KEYKE

Step 3: Convert the letters into numerical values.

Using the alphabet where A = 0, B = 1, ..., Z = 25:

H = 7

E = 4

L = 11

L = 11

O = 14

Keyword values:

K = 10

$$E = 4$$

$$Y = 24$$

$$K = 10$$

$$E = 4$$

Step 4: Apply the Vigenère encryption formula.

$$\text{Cipher} = (\text{Plaintext} + \text{Keyword}) \bmod 26$$

For each letter:

$$H + K = 7 + 10 = 17 = R$$

$$E + E = 4 + 4 = 8 = I$$

$$L + Y = 11 + 24 = 35 \bmod 26 = 9 = J$$

$$L + K = 11 + 10 = 21 = V$$

$$O + E = 14 + 4 = 18 = S$$

Therefore, the encrypted message is:

RIJVS

b. Explain how the repeating keyword influences each character of the ciphertext.

The repeating keyword is the main feature that distinguishes the Vigenère Cipher from the Caesar Cipher. Instead of applying one fixed shift to every letter, each letter of the plaintext is shifted according to the corresponding letter of the keyword.

Since the keyword **KEY** repeats as **KEYKE**, each plaintext letter is encrypted using a different shift value. This means that identical plaintext letters appearing in different positions may produce different ciphertext letters.

For example, the letter **L** appears twice in the plaintext, but because it is paired with different keyword letters (**Y** and **K**), it becomes **J** in one position and **V** in another.

This significantly increases security because it makes frequency analysis much more difficult than in monoalphabetic ciphers. Although stronger than the Caesar Cipher, the Vigenère Cipher can still be broken using methods such as the Kasiski Examination or Index of Coincidence if enough ciphertext is available.

4. Scenario: Suspicious Digital Asset Investigation

During a digital forensic investigation, analysts suspect that a company logo contains hidden confidential information. The image appears completely normal during visual inspection.

a. Analyze how investigators could detect the presence of hidden data within the image.

Investigators use a process called **steganalysis** to determine whether an image contains hidden information.

One common method is **Least Significant Bit (LSB) analysis**. In many steganographic techniques, secret information is hidden by modifying the least significant bits of pixel values. These tiny changes are invisible to the human eye but can be detected using specialized software.

Another method is **statistical analysis**. Investigators compare the statistical properties of the image with those of normal images. Hidden data often alters the natural distribution of pixel values, allowing forensic tools to detect anomalies.

Metadata analysis is also useful. Investigators examine image metadata to identify unusual modifications, suspicious timestamps, or embedded information that may indicate steganography.

Specialized steganalysis tools such as **StegExpose**, **StegSolve**, and **zsteg** can automatically scan image files for hidden content and identify commonly used embedding techniques.

Investigators may also compare the suspected image with the original version. Differences between the two images may reveal the locations where hidden information has been embedded.

b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

The first technique is **encrypting the secret message before embedding it into the image**. Even if investigators successfully extract the hidden data, they will still require the encryption key to understand the message. Combining cryptography with steganography provides two layers of security.

The second technique is **randomized embedding using a secret key**. Instead of hiding data sequentially in image pixels, the embedding locations are selected randomly using a secret key. Without knowledge of this key, it becomes extremely difficult for an attacker to locate or extract the hidden information.

Additional methods such as compressing the message before embedding it, selecting high-resolution cover images, and using adaptive steganographic algorithms can further improve security and reduce the likelihood of detection.

5. Scenario: Legacy Encryption in Corporate Communication

A company continues to use a legacy transposition cipher with three levels of arrangement. A received message appears as:

WECRLTEERDSOEFEAOCAIVDEN

a. Determine the most probable original message by reversing the encryption process.

The ciphertext is encrypted using a **3-Rail Fence Cipher**, a classical transposition cipher.

Unlike substitution ciphers, the Rail Fence Cipher does not replace letters. Instead, it rearranges their positions in a zigzag pattern across multiple rails.

By reconstructing the zigzag arrangement and reading the letters in the correct order, the original message becomes:

WE ARE DISCOVERED FLEE AT ONCE

This is a well-known example used to demonstrate the Rail Fence Cipher.

b. Critically evaluate why such a method is vulnerable in modern communication systems.

Although the Rail Fence Cipher was useful historically, it provides very weak security by modern standards.

Since the letters themselves remain unchanged, attackers can still perform frequency analysis on the ciphertext. The only secret is the number of rails, which is usually a very small value.

Modern computers can rapidly test every possible rail arrangement using brute-force techniques. Once the correct number of rails is found, the original message is immediately recovered.

The cipher also provides no protection against known-plaintext attacks, chosen-plaintext attacks, or automated cryptanalysis. Because it lacks a cryptographic key and uses only letter rearrangement, it cannot provide confidentiality for sensitive corporate communications.

Consequently, the Rail Fence Cipher is considered obsolete and should not be used in modern information security systems.

c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

A practical improvement would be to replace the Rail Fence Cipher with the **Advanced Encryption Standard (AES)**.

AES is a modern symmetric encryption algorithm that supports key lengths of 128, 192, and 256 bits. It performs multiple rounds of substitution, permutation, and mathematical transformations to produce highly secure ciphertext.

Using AES offers several advantages. It is resistant to brute-force attacks, widely accepted as an international encryption standard, and used by governments, banks, healthcare organizations, and technology companies worldwide. Even if an attacker intercepts an AES-encrypted message, recovering the plaintext without the correct key is computationally infeasible.

For even greater security, organizations can combine AES encryption with secure key management, digital signatures, authentication mechanisms, and steganography. These measures ensure confidentiality, integrity, and authenticity of sensitive communications.

Conclusion

Classical cryptographic techniques such as the Caesar Cipher, Atbash Cipher, Vigenère Cipher, and Rail Fence Cipher illustrate the fundamental concepts of encryption, decryption, substitution, and transposition. These methods played a significant role in the historical development of secure communication and remain valuable educational tools for understanding the principles of cryptography. However, they possess several weaknesses, including susceptibility to brute-force attacks, frequency analysis, and pattern-based cryptanalysis.

Modern communication systems require much stronger protection. Contemporary encryption algorithms such as AES, when combined with secure key management, authentication mechanisms, and steganography, provide a much higher level of confidentiality and resistance against modern cyber threats. Therefore, while classical ciphers remain important for academic study, they should not be relied upon for securing sensitive information in today's digital environment.